

Last Class

1) Hierarchical Clustering
↳ Agglomerative Clustering

2) Agglomerative Clustering

- ↳ Dendrogram
- ↳ Types of distances
- ↳ Proximity Matrix
- ↳ sklearn implementation

Today's class

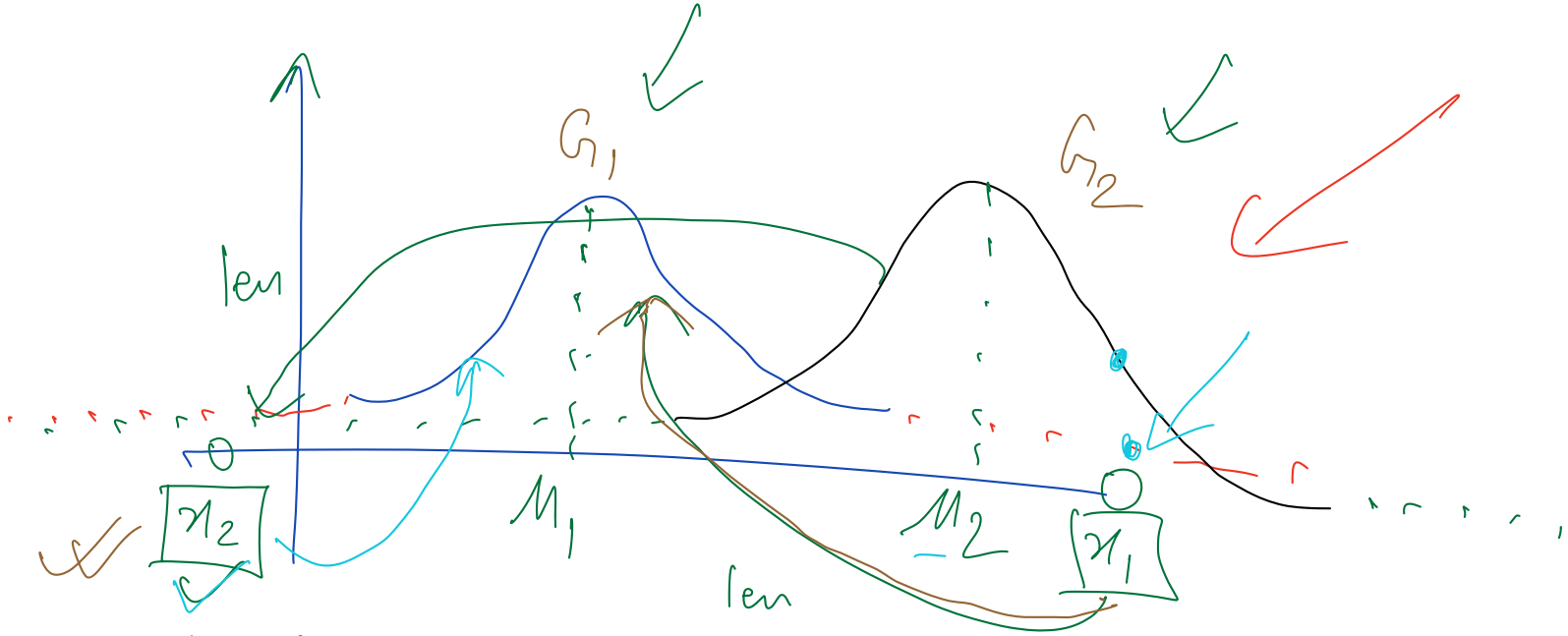
- 1) Intro to Gaussian Mixture Models (GMM)
- 2) Extending GMMs to multi-dimension
- 3) GMM Algo & implementation

$$e^{-\left(\frac{x - \mu}{\sigma}\right)^2}$$

↓

-∞ to +∞

(Annotations: "arg" points to the exponent, "std" points to the denominator σ)



$$\underline{\text{pdf}(x_2)} = \text{pdf}_{G_1}(x_2) + \text{pdf}_{G_2}(x_2)$$

↓
high

↓
low

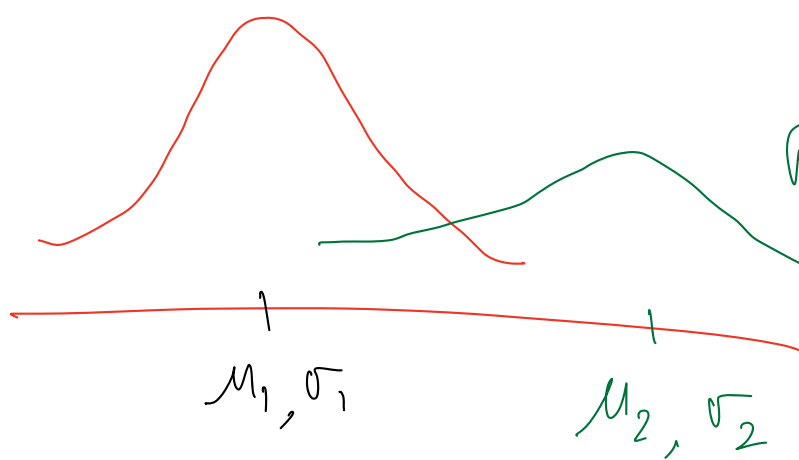
$$\underline{\text{pdf}(x_1)} = \underbrace{\text{pdf}_{G_1}(x_1)}_{\text{len}} + \underbrace{\text{pdf}_{G_2}(x_1)}_{\text{more}}$$

k-Means

	0	1	2	3	...	k-1	→ cluster no.s
x_1	0	1	0	0	0	0	
x_2	0	0	1	0	0	0	
x_3	1			...			
x_4	1						
$\vdots$							
x_N	0	0	0	0	0	1	

$N \times k$

	0	1	2	→ clusters
x_1	0.98	0.01	0.01	→ 1.0
x_2	0.1	0.5	0.4	→ 1.0
x_3	0.34	0.33	0.33	→ 1.0

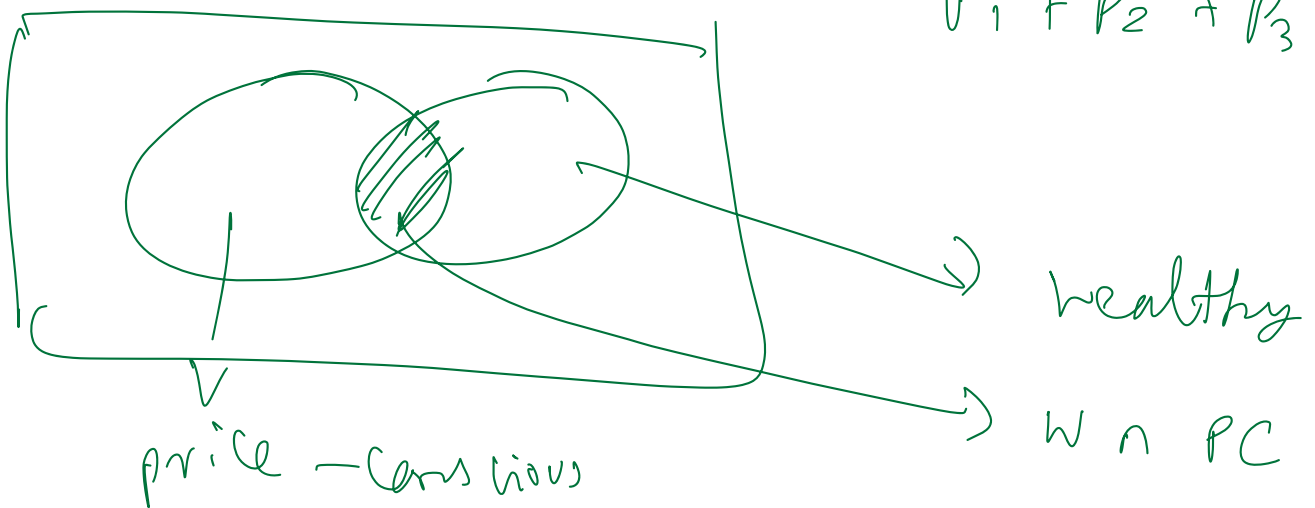


$$P(x_1 | G_1) = p_1$$

$$P(x_1 | G_2) = p_2$$

$$P(x_p | G_3) = p_3$$

$$p_1 + p_2 + p_3 = 1$$



1D → n values

μ_1

2D space

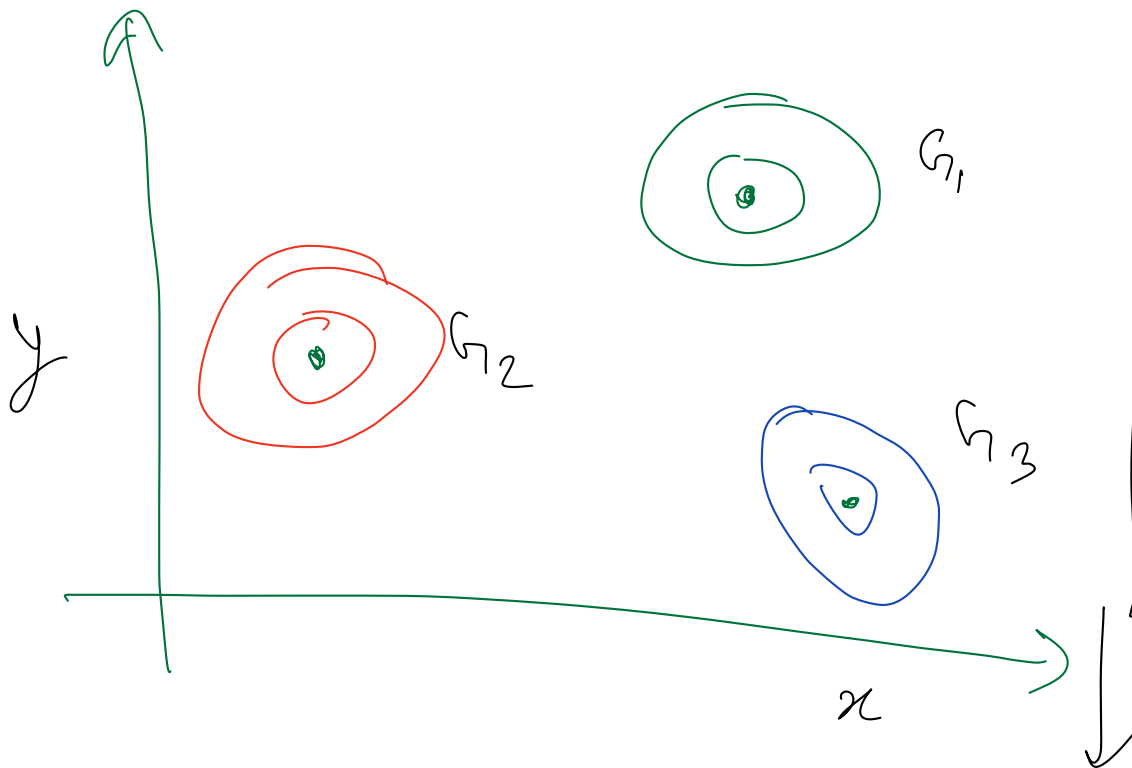
x, y

$$\rightarrow \begin{bmatrix} \mu_x, \mu_y \end{bmatrix}$$

2x2 matrix Covariance

$$\boxed{\text{Cov}(x, y)} = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{bmatrix}$$

μ



μ_1, μ_2, μ_3
 $\sigma_1, \sigma_2, \sigma_3$

$$\begin{aligned} \mu_1 &= [\mu_{1x}, \mu_{1y}] \\ \mu_2 &= [\mu_{2x}, \mu_{2y}] \\ \mu_3 &= [\mu_{3x}, \mu_{3y}] \end{aligned}$$

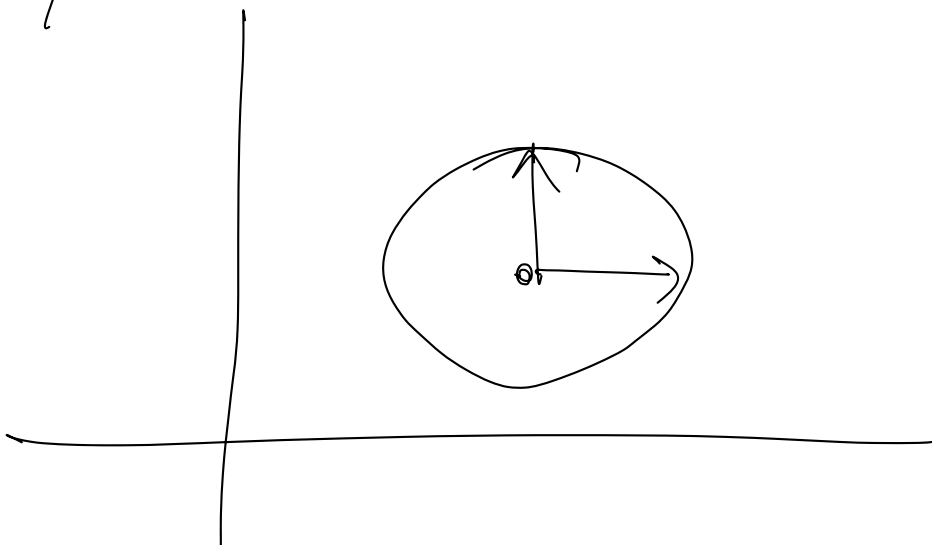
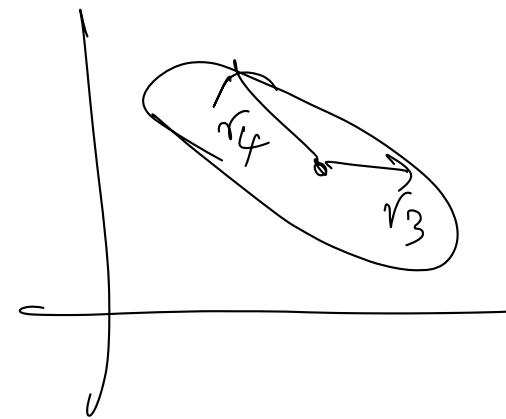
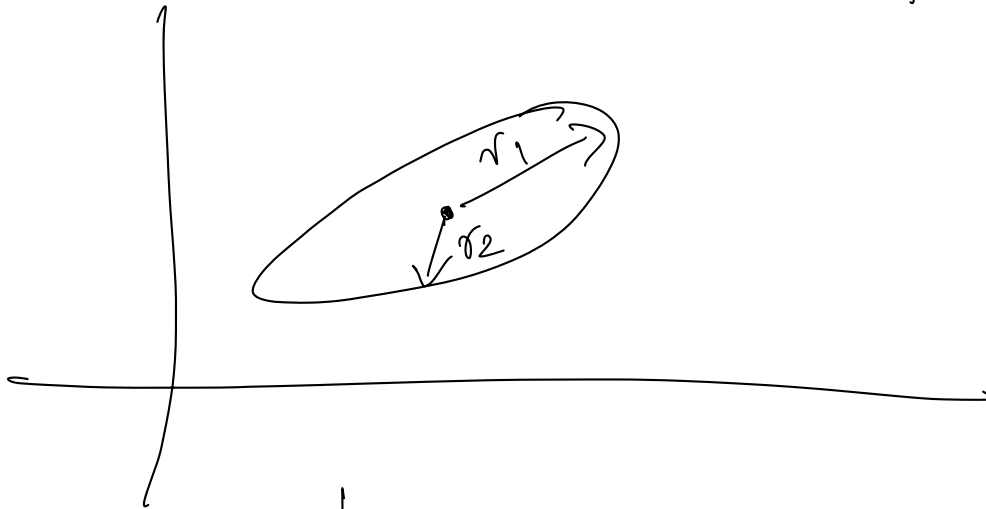
$$\text{Cov}_{G_1} = \begin{bmatrix} & \end{bmatrix}_{2 \times 2}$$

$$\text{Cov}_{G_2} = \begin{bmatrix} & \end{bmatrix}$$

$$\text{Cov}_{G_3} = \begin{bmatrix} & \end{bmatrix}$$

circle: $ax^2 + by^2 + dx + ey + f = 0$

ellipse: $ax^2 + by^2 + f \begin{matrix} \boxed{cxy} \\ \text{ray} \end{matrix} + dx + ey + f = 0$



3 - dim space, 4 Gaussians
 $\downarrow$
 M, cov

$$\begin{array}{c}
 \left[\begin{array}{cc}
 \mu_1, \text{cov}_1 \\
 \mu_2, \text{cov}_2 \\
 \mu_3, \text{cov}_3 \\
 \mu_4, \text{cov}_4
 \end{array} \right]
 \end{array}
 \begin{array}{c}
 \downarrow \\
 3 \times 1 \\
 \text{row vector}
 \end{array}
 \begin{array}{c}
 \downarrow \\
 3 \times 3
 \end{array}$$

same dim

$$4 \times \left[3 \times 1, 3 \times 3 \right]$$

$$\begin{array}{c}
 [x_1, y_1] \rightarrow \text{2D space} \\
 \underline{1 \times 2}
 \end{array}$$

$$x = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \underline{2 \times 1}$$

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} 2 \times 1$$

$$x - \mu = \begin{bmatrix} x_1 - \mu_1 \\ y_1 - \mu_2 \end{bmatrix} 2 \times 1$$

$$(x - \mu)^T \rightarrow 1 \times 2 \quad \checkmark$$

$$\Sigma \rightarrow 2 \times 2$$

$$x - \mu \rightarrow 2 \times 1$$

$$(1 \times 2) \times (2 \times 2) \times (2 \times 1)$$

$\hookrightarrow$

1×2

$\nwarrow$

$[1 \times 1]$

$\downarrow$
scalar value

k - Means

1) centroid initialisation $\rightarrow$ centroid rec

num-iterations $\rightarrow$ loop

2) cluster assignment

3) centroid update / re-computation

GMM

1) initialisation of Centroid vectors,
COV matrices

num-iterations $\rightarrow$ loop

$K=4$

2) probability computation

$\begin{bmatrix} 0.1 & 0.5 & 0.2 & 0.2 \\ 0 & 1 & 2 & 3 \end{bmatrix}$

Centroid + covariance update

cluster 1

cluster 2

x_1

$\boxed{C_1}$

x_1

p_1

p'_1

x_2

p_2

p'_2

x_3

p_3

p'_3

$\vdots$

x_N

p_N

p'_N

$$C'_1 = p_1 \times x_1 + p_2 \times x_2$$

$$+ \dots + p_N \times x_N$$

$$p_1 + p_2 + \dots + p_N$$

$$\boxed{C'_2} = \frac{p'_1 \times x_1 + p'_2 \times x_2 + \dots + p'_N \times x_N}{p'_1 + p'_2 + \dots + p'_N}$$

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \quad \text{cov} = (x - \mu)(x - \mu)^T$$

2×1

$$\begin{bmatrix} x_1 - \mu_1 \\ y_1 - \mu_2 \end{bmatrix} \times \begin{bmatrix} x_1 - \mu_1, y_1 - \mu_2 \end{bmatrix}$$

2×1

$\frac{1}{N}$

$\sum$

$$\begin{bmatrix} (x_1 - \mu_1)^2 & (x_1 - \mu_1)(y_1 - \mu_2) \\ (x_1 - \mu_1)(y_1 - \mu_2) & (y_1 - \mu_2)^2 \end{bmatrix}$$

cov

$$= p_1 \times \text{cov}(x_1) + p_2 \times \text{cov}(x_2) + \dots$$

$$p_1 + p_2 + p_3 + \dots + p_n$$

$$\sum d(x_i, c_j)$$

K-Means Loss fn

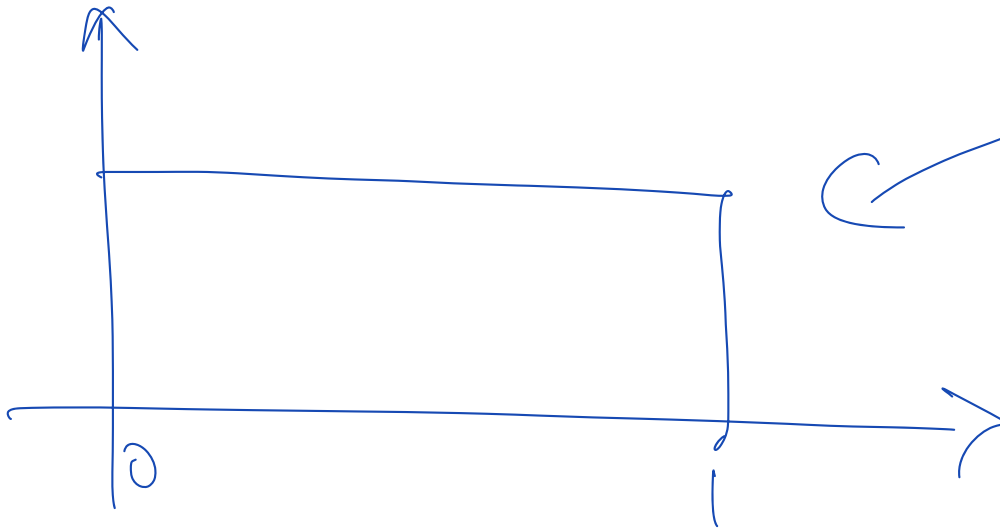
$\rightarrow$

$$\sum_{i=1}^N$$

$$|x_i - c_j|^2$$

$$x_i \in c_j$$

Within Cluster Sum of Squares



10 values

0.1

0.6

0.3

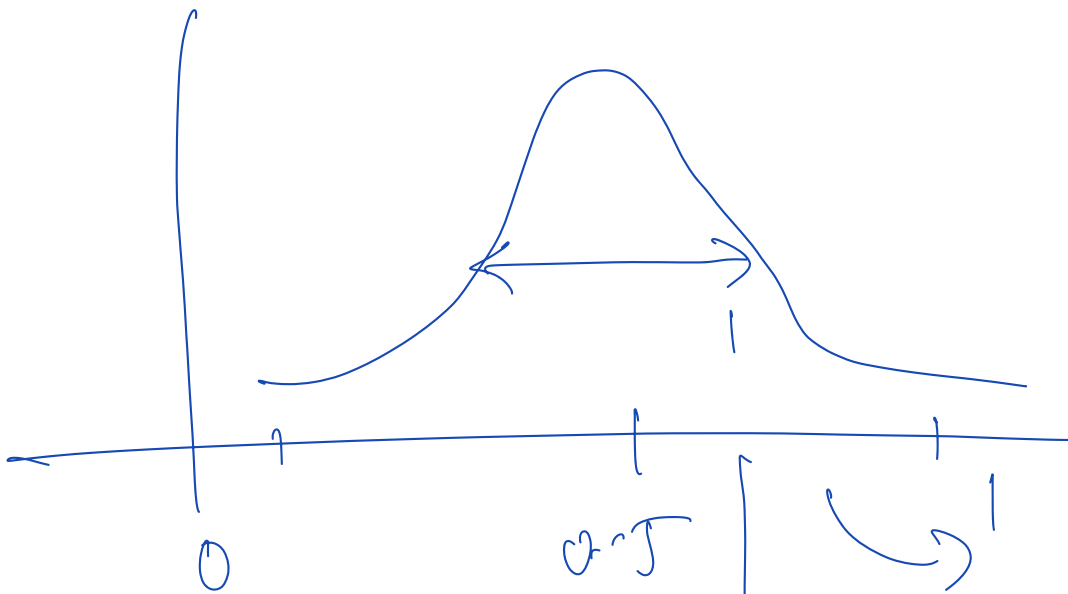
0.15

0.1

...

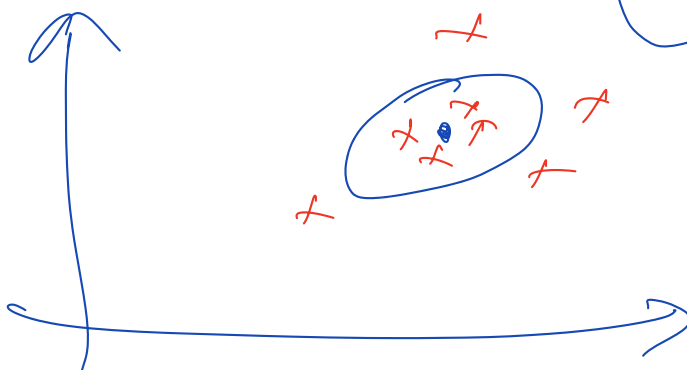
0.1

0.2



0.5

1



0.51
0.49
0.5
0.52
0.6

0.3